

- Hide menu
- Week 3: Image Sensing
- Reading: Week 3 Lecture Handout

30 min
- Ungraded Plugin: 3.1 Overview of Imaging Sensing

3 min
- Practice Quiz: 3.1 Overview of Image Sensing Self-check Quiz

Started
- Ungraded Plugin: 3.2 A Brief History of Imaging

16 min
- Practice Quiz: 3.2 A Brief History of Imaging Self-check Quiz

Started
- Reading: 3.3 Video Correction

10 min
- Ungraded Plugin: 3.3 Types of Image Sensors

11 min
- Practice Quiz: 3.3 Types of Image Sensors Self-check Quiz

10 min
- Reading: 3.4 Video Correction

10 min
- Ungraded Plugin: 3.4 Resolution, Noise, Dynamic Range

14 min
- Practice Quiz: 3.4 Resolution, Noise and Dynamic Range Self-check Quiz

15 min
- Ungraded Plugin: 3.5 Sensing Color

19 min
- Reading: 3.5 Sensing Color Supplementary Reading

10 min
- Practice Quiz: 3.5 Sensing Color Self-check Quiz

10 min
- Ungraded Plugin: 3.6 Camera Response and HDR Imaging

17 min
- Practice Quiz: 3.6 Camera Response and HDR Imaging Self-check Quiz

10 min
- Ungraded Plugin: 3.7 Nature's Image Sensors

8 min
- Discussion Prompt: Week 3 Designing a Camera Lens

10 min
- Discussion Prompt: Week 3 Questions and Feedback
- Week 3 Quiz

🎉 Congratulations! You passed!

Grade received 100%

To pass 50% or higher

Go to next item

3.3 Types of Image Sensors Self-check Quiz

1 / 1 point

1. Which property of Silicon atoms is leveraged in sensors?

Releases electron flux greater than incoming photon flux

Releases electron flux equal to incoming photon flux

Releases electron flux lesser than incoming photon flux

Correct

Please refer to course video for explanation.

Submit your assignment

Try again

2. What occurs during bucket brigading in CCD sensors?

Photons are transferred from one row to the next row.

Electrons are transferred from one row to the next row.

Each pixel has its own circuit to convert electrons to voltage.

Correct

Please refer to lecture video.

Like

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3. CMOS sensors have a higher frame rate than CCD sensors

True

False

Correct

Please refer to lecture video.

4. Why is the exposed light-sensitive area smaller in CMOS sensors than in CCD sensors?

The additional circuitry takes up space.

They require more power.

CCD has better circuitry for electron to voltage conversion.

Correct

Please refer to course video for explanation

Your grade

100%

1 / 1 point

View Feedback

We keep your highest score

https://www.coursera.org/learn/cameraandimaging/quiz/Ky0x3-3-types-of-image-sensors-self-check-quiz/attemptedirectToCovertrue

1/1